

Monoalphabetic Cipher Encryption and Decryption

1A

INPUT:-

```
# Monoalphabetic Cipher

plain = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
cipher = "QWERTYUIOPASDFGHJKLZXCVBNM"

# Encryption
def encrypt(text):
    result = ""
    for ch in text.upper():
        if ch in plain:
            index = plain.index(ch)
            result += cipher[index]
        else:
            result += ch
    return result

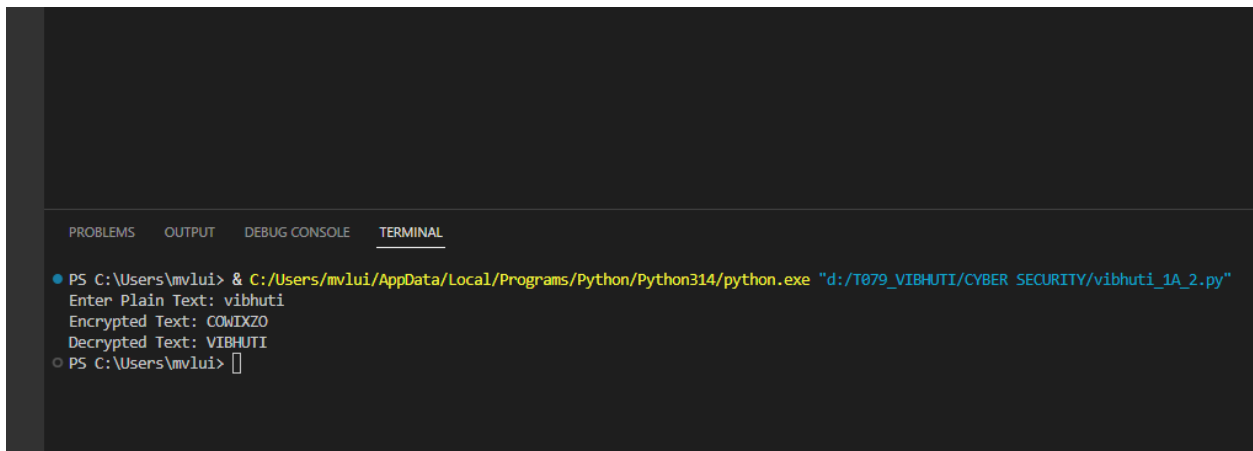
# Decryption
def decrypt(text):
    result = ""
    for ch in text.upper():
        if ch in cipher:
            index = cipher.index(ch)
            result += plain[index]
        else:
            result += ch
    return result

# Main Program
message = input("Enter Plain Text: ")

encrypted = encrypt(message)
print("Encrypted Text:", encrypted)

decrypted = decrypt(encrypted)
print("Decrypted Text:", decrypted)
```

OUTPUT:-



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PS C:\Users\mvlui> & C:/Users/mvlui/AppData/Local/Programs/Python/Python314/python.exe "d:/T079_VIBHUTI/CYBER SECURITY/vibhuti_1A_2.py"
Enter Plain Text: vibhuti
Encrypted Text: COWIXZO
Decrypted Text: VIBHUTI
PS C:\Users\mvlui>
```

GUI:

INPUT:-

```
from tkinter import *

# Plain and Cipher Alphabets
plain_alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
cipher_alphabet = "QWERTYUIOPASDFGHJKLZXCVBNM"

# Encryption Function
def encrypt():
    plaintext = entry.get().upper()
    ciphertext = ""

    for ch in plaintext:
        if ch in plain_alphabet:
            index = plain_alphabet.index(ch)
            ciphertext += cipher_alphabet[index]
        else:
            ciphertext += ch

    result.config(text="Encrypted Text: " + ciphertext)

# Decryption Function
def decrypt():
    ciphertext = entry.get().upper()
    plaintext = ""

    for ch in ciphertext:
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        if ch in cipher_alphabet:
            index = cipher_alphabet.index(ch)
            plaintext += plain_alphabet[index]
        else:
            plaintext += ch

    result.config(text="Decrypted Text: " + plaintext)

# Clear Function
def clear():
    entry.delete(0, END)
    result.config(text="Result will appear here")

# GUI Window
root = Tk()
root.title("Monoalphabetic Cipher")
root.geometry("500x300")
root.configure(bg="lightblue")

# Heading
Label(root,
      text="Monoalphabetic Cipher",
      font=("Arial", 18, "bold"),
      bg="lightblue").pack(pady=10)

# Input
Label(root,
      text="Enter Text:",
      font=("Arial", 12),
      bg="lightblue").pack()

entry = Entry(root, font=("Arial", 14), width=35)
entry.pack(pady=10)

# Buttons
Button(root,
      text="Encrypt",
      font=("Arial", 12),
      width=12,
      bg="green",
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        fg="white",
        command=encrypt).pack(pady=5)

Button(root,
        text="Decrypt",
        font=("Arial", 12),
        width=12,
        bg="blue",
        fg="white",
        command=decrypt).pack(pady=5)

Button(root,
        text="Clear",
        font=("Arial", 12),
        width=12,
        bg="red",
        fg="white",
        command=clear).pack(pady=5)

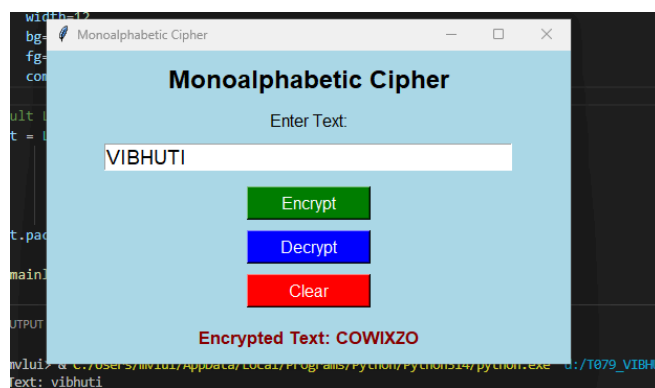
# Result Label
result = Label(root,
                text="Result will appear here",
                font=("Arial", 13, "bold"),
                bg="lightblue",
                fg="darkred")

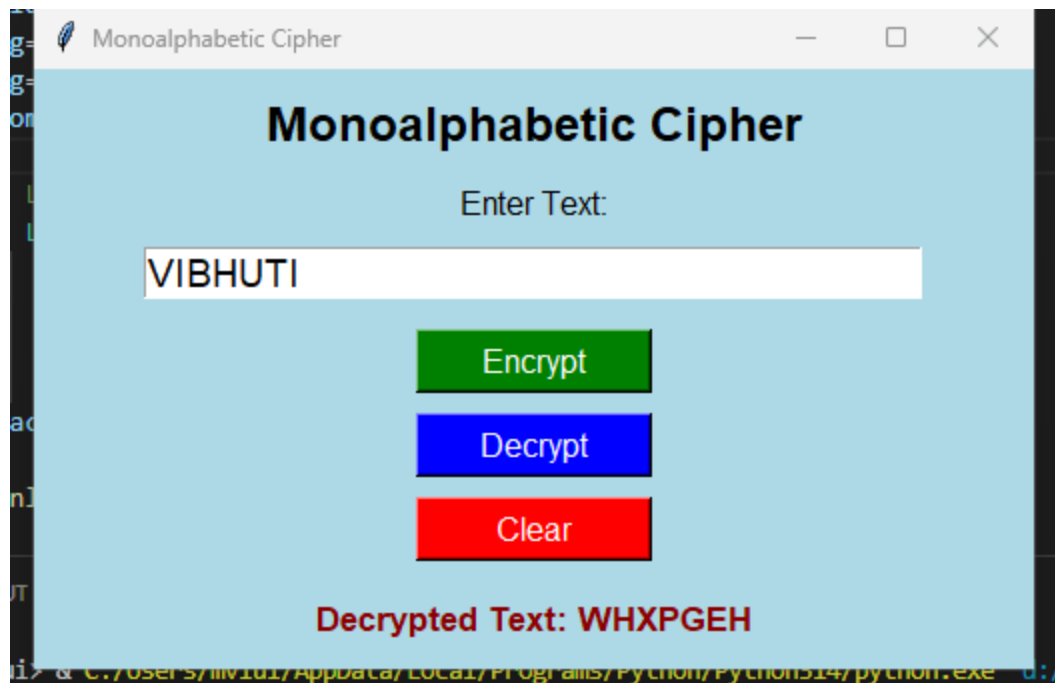
result.pack(pady=15)

root.mainloop()

```

OUTPUT:-





Columnar Transposition Cipher

1B:-

INPUT:-

```
import math

def encrypt(message, key):
    message = message.replace(" ", "").upper()
    key = key.upper()

    col = len(key)
    row = math.ceil(len(message) / col)

    message += "X" * (row * col - len(message))

    matrix = []
    index = 0

    for i in range(row):
        matrix.append(list(message[index:index + col]))
        index += col

    order = sorted(range(col), key=lambda k: key[k])
```

```

cipher = ""

for c in order:
    for r in range(row):
        cipher += matrix[r][c]

return cipher

def decrypt(cipher, key):
    cipher = cipher.replace(" ", "").upper()
    key = key.upper()

    col = len(key)
    row = math.ceil(len(cipher) / col)

    matrix = [[' ' for _ in range(col)] for _ in range(row)]

    order = sorted(range(col), key=lambda k: key[k])

    index = 0

    for c in order:
        for r in range(row):
            matrix[r][c] = cipher[index]
            index += 1

    plain = ""

    for r in range(row):
        for c in range(col):
            plain += matrix[r][c]

    return plain.rstrip("X")

# Main Program
message = input("Enter Message: ")
key = input("Enter Key: ")

```

```

cipher = encrypt(message, key)
print("Encrypted Text:", cipher)

plain = decrypt(cipher, key)
print("Decrypted Text:", plain)

```

OUTPUT:-

```

● PS C:\Users\mvlui> & C:/Users/mvlui/AppData/Local/Programs/Python/Python314/python.exe "d:/T079_VIBHUTI/CY
● PS C:\Users\mvlui> & C:/Users/mvlui/AppData/Local/Programs/Python/Python314/python.exe "d:/T079_VIBHUTI/CY
● PS C:\Users\mvlui> & C:/Users/mvlui/AppData/Local/Programs/Python/Python314/python.exe "d:/T079_VIBHUTI/CY
Enter Message: VIBHUTI
Enter Key: 2
Encrypted Text: VIBHUTI
Decrypted Text: VIBHUTI
○ PS C:\Users\mvlui> 

```

1B GUI

INPUT:-

```

from tkinter import *
import math

def encrypt():
    message = message_entry.get().replace(" ", "").upper()
    key = key_entry.get().upper()

    if message == "" or key == "":
        result_label.config(text="Enter Message and Key")
        return

    col = len(key)
    row = math.ceil(len(message) / col)

    message += "X" * (row * col - len(message))

    matrix = []
    index = 0
    for i in range(row):
        matrix.append(list(message[index:index + col]))
        index += col

    order = sorted(range(col), key=lambda k: key[k])

```

```

cipher = ""
for c in order:
    for r in range(row):
        cipher += matrix[r][c]

result_label.config(text="Encrypted Text:\n" + cipher)

def decrypt():
    cipher = message_entry.get().replace(" ", "").upper()
    key = key_entry.get().upper()

    if cipher == "" or key == "":
        result_label.config(text="Enter Cipher Text and Key")
        return

    col = len(key)
    row = math.ceil(len(cipher) / col)

    matrix = [['' for _ in range(col)] for _ in range(row)]

    order = sorted(range(col), key=lambda k: key[k])

    index = 0
    for c in order:
        for r in range(row):
            matrix[r][c] = cipher[index]
            index += 1

    plain = ""
    for r in range(row):
        for c in range(col):
            plain += matrix[r][c]

    plain = plain.rstrip("X")

    result_label.config(text="Decrypted Text:\n" + plain)

```

```

def clear():
    message_entry.delete(0, END)
    key_entry.delete(0, END)
    result_label.config(text="")

root = Tk()
root.title("Columnar Transposition Cipher")
root.geometry("500x350")

Label(root, text="Columnar Transposition Cipher", font=("Arial", 16,
"bold")).pack(pady=10)

Label(root, text="Enter Message").pack()
message_entry = Entry(root, width=40)
message_entry.pack(pady=5)

Label(root, text="Enter Key").pack()
key_entry = Entry(root, width=20)
key_entry.pack(pady=5)

Button(root, text="Encrypt", width=12, command=encrypt).pack(pady=5)
Button(root, text="Decrypt", width=12, command=decrypt).pack(pady=5)
Button(root, text="Clear", width=12, command=clear).pack(pady=5)

result_label = Label(root, text="", font=("Arial", 12))
result_label.pack(pady=15)

root.mainloop()

```

OUTPUT:-

